Sharp thresholds for five-colour cube assemblies

Jason Davies

November 4, 2026

Editorial integration draft: companion mathematical appendix under consolidation.

Abstract

We determine the least pool sizes guaranteeing a frame or a complete $a \times b \times c$ box from unit cubes whose faces use all five colours of a fixed palette, for every $a, b, c \geq 2$. Proper rotations are allowed; the target has monochromatic exterior faces using all five colours. The frame threshold is $\max\{15, 4(a+b+c)-16, 2(a+b+c)+1\}$, and the complete-box threshold is $\max\{abc, 15, 2(a+b+c)+1\}$. Thus the frame threshold depends only on the sum of the side lengths, and only three complete-box shapes can require spare sources. The proof combines an elementary fifteen-cube corner theorem with Hall arguments and a uniform three-variety obstruction. The corner proof uses directed cuts, repeated-colour completions and geometric case analysis; its longer cases are proved in a companion mathematical appendix. It completes the five-colour frame column left open by Berkove, Eickemeyer and Schulman (2018). The same threshold statements have a separate Lean verification, using a certificate-based proof of the corner theorem. A fixed-size flow formulation constructs assemblies in time linear in the explicit input and output sizes. A secondary result, using additional finite checks, determines corner thresholds when the number of source varieties is bounded.

AI transparency. The research, code and manuscript were developed using GPT-6-Astra (OpenAI), through Codex, under the author's direction. This working manuscript has not received independent human peer review; Section 9 states the formal verification scope.

1 Introduction

MacMahon's coloured-cube puzzles [1] ask when a collection of face-coloured cubes can form a larger cube with monochromatic exterior faces. Here every source and target uses all five colours of a fixed palette. Proper rotations identify varieties; reflections are not allowed as moves. Collections may contain repeated varieties, and touching interior faces need not have equal colours. For $a, b, c \geq 2$, a *frame* consists of the eight corners and the noncorner positions on the twelve edges of an $a \times b \times c$ box. It has $L = 4(a+b+c) - 16$ positions. Write $F_5(a, b, c)$ for the least pool size guaranteeing a frame and $G_5(a, b, c)$ for the least pool size guaranteeing a complete box. An assembly may use a subset of the pool; its target colouring may depend on the collection.

Theorem 1 (All cuboid thresholds)**.** *For every integer $a, b, c \geq 2$,*

$$F_5(a, b, c) = \max\{15, 4(a+b+c) - 16, 2(a+b+c) + 1\},$$
$$G_5(a, b, c) = \max\{abc, 15, 2(a+b+c) + 1\}.$$

The frame threshold depends only on the sum $s = a+b+c$: it is 15 for $s = 6, 7$, 17 for $s = 8$, and the frame's own cardinality $4s - 16$ for $s \geq 9$. Up to permuting sides, only three complete boxes can require spare sources: $2 \times 2 \times 2$, $2 \times 2 \times 3$, and $2 \times 2 \times 4$, whose thresholds are $15, 15, 17$. The distinction between the two thresholds matters: a $2 \times 3 \times 3$ frame has sixteen positions and needs seventeen available sources, whereas its complete box uses any eighteen sources without

Box	s	Frame positions	F_5	Volume	G_5
$2\times2\times2$	6	8	**15**	8	**15**
$2\times2\times3$	7	12	**15**	12	**15**
$2\times2\times4$	8	16	**17**	16	**17**
$2\times3\times3$	8	16	**17**	18	18
$2\times2\times5$	9	20	20	20	20
$2\times3\times4$	9	20	20	24	24
$3\times3\times3$	9	20	20	27	27

Table 1: Thresholds for the smallest boxes. A bold entry exceeds the number of positions: some collections with no spare sources fail.

spares. Table 1 lists the smallest shapes; from $s = 9$ on, both thresholds equal the number of positions.

For ordinary cubes, abbreviate $F_5(n) = F_5(n,n,n)$ and $G_5(n) = G_5(n,n,n)$. The classification immediately gives the following.

Corollary 2 (Ordinary cubes)**.**

$$F_5(n) = \begin{cases} 15, & n = 2, \\ 12n - 16, & n \geq 3, \end{cases} \qquad G_5(n) = \max\{n^3, 15\} \quad (n \geq 2).$$

In particular, any twenty sources form the frame of a $3\times3\times3$ *cube, and any 27 form a complete* $3\times3\times3$ *cube.*

Berkove, Eickemeyer and Schulman [2, Section 8, Table 7] record frame thresholds for palettes of two to six colours, leaving the five-colour column open. Table 2 inserts the values proved here; the cuboid classification goes beyond that table.

Side length	2 colours	3 colours	4 colours	**5 colours**	6 colours
$n = 2$	11	23	11	**15**	24
$n = 3$	20	35	20	**20**	24
$n \geq 4$	$12n - 16$	$16n - 17$	$12n - 16$	$\mathbf{12n - 16}$	$12n - 16$

Table 2: Cube-frame thresholds. The five-colour column is proved here; other columns are as recorded in [2, Table 7].

We first prove the corner threshold independently of the extensions. Write $N_5 = F_5(2) = G_5(2)$.

Theorem 3 (Corner threshold)**.** $N_5 = 15$.

An elementary fourteen-cube obstruction proves the lower bound. For the upper bound, adjacent corner classes become directed cuts on five colours, while opposite-repeat varieties become bidirected four-cycles. Small matching constructions and repeated-colour arguments constrain a hypothetical failure. Compression of disjoint cuts reduces the two-opposite case to three geometric support families. Section 5 combines these ingredients into a direct fifteen-source proof; all necessary longer cases are included in the companion mathematical appendix, submitted as part of the proof rather than as optional computational material.

Section 3 establishes the shared geometry, including the eight-plus-three missing-pair lemma independently of the corner theorem. Section 6 extends the corner input to all cuboids. A uniform three-variety obstruction explains the two sixteen-position exceptions. Section 7 gives a compressed flow algorithm, and Section 8 treats bounded numbers of source varieties using separate finite geometric checks. The elementary-proof claim concerns the main threshold classification. Section 9 distinguishes that argument from the existing formal verification.

The side-length restriction is substantive. With a side of length one, a source can meet two opposite exterior faces and expose four or more faces at once. The corner-and-edge model and the completion by arbitrary unused cubes then need replacement.

The accompanying repository contains the certificates, programs, geometric audits and Lean sources.[1]

The companion paper [5] considers four-colour sources and targets using exactly three colours. It gives the sharp frame threshold $16n - 3$ for every side length $n \geq 2$, with a classification of the largest failing collections. The two developments share the palette-independent geometry in the `colour-cubes-core` library [6]; neither threshold theorem is used to prove the other.

2 Geometry and Hall deficiencies

Represent a cube by its colours on the faces $(+x, -x, +y, -y, +z, -z)$ in a right-handed coordinate frame. A five-colour cube has one repeated colour. There are 75 varieties: for each repeated colour, three have its two faces opposite and twelve have them adjacent. In the opposite case, the other four colours form two opposite pairs, giving three possibilities. In the adjacent case, fix the two equal faces; the 24 orders of the other colours are identified in pairs by the rotation exchanging those equal faces. Thus there are 15 opposite-repeat varieties and 60 adjacent-repeat varieties. Each class is one orbit under colour permutations.

A corner type is the ordered triple of its incident face colours, modulo cyclic permutation. Order the outward face normals positively, equivalently read the triple counterclockwise from outside along the body diagonal. This reverses the clockwise convention in [2], consistently on every cube, and therefore preserves all matching and containment relations. Repeated occurrences of a corner type are retained as distinct target positions.

Lemma 4 (Corner–rotation correspondence)**.** *A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type at some corner.*

Proof. A proper rotation preserves the orientation of a corner frame, so it induces a cyclic permutation of its three outward faces. Conversely, any two positively oriented corner frames, with a prescribed cyclic alignment, are related by a determinant-positive signed permutation of the coordinate axes. This is a proper rotation of the cube. □

The palette-independent Lean version of this lemma is reused from the shared `colour-cubes-core` library [6].

Fix a target T. Form a bipartite graph whose left vertices are its eight corner positions and whose right vertices are the individual source copies. A copy is adjacent to a position when it fits there. A matching covering the eight positions is exactly a solution: each chosen copy can be rotated independently, since interior faces are unrestricted. By Hall's theorem [3], failure is equivalent to a nonempty set D of target positions having fewer than $|D|$ neighbouring copies.

Index the varieties by $i = 1, \ldots, 75$, and let x_i be their multiplicities. For a set S of varieties, write $x(S) = \sum_{i \in S} x_i$. Let $N_T(D)$ be the varieties fitting at least one position of D. Then failure of T is equivalent to the existence of a cut

$$x(N_T(D)) \leq |D| - 1, \qquad \varnothing \neq D \subseteq \{1, \ldots, 8\}. \tag{1}$$

There are 255 cuts per target, hence 19,125 cuts altogether.

[1] https://github.com/jasondavies/five-colour-cubes. A fixed review archive includes source checksums, the base Git revision and bundled pinned project dependencies.

3 Shared geometric and matching tools

The corner and frame arguments share the following geometric and matching tools. All statements in this section are independent of the corner threshold. Lemmas 6 and 7 state the shape-independent facts used in the directional bounds of Section 6. The following form of Hall's theorem isolates the matching step.

Lemma 5 (Hall criterion with a matched subset)**.** *Consider L target positions and $N \geq L$ source copies. Suppose every source misses at most d positions. If a set P_0 of positions can be matched to distinct copies, and every position outside P_0 has at least d neighbours, then all L positions can be matched.*

Proof. For a position set $D \subseteq P_0$, the given matching supplies $|D|$ neighbours. If $D \not\subseteq P_0$ and $|D| \leq d$, a position in $D \setminus P_0$ alone has at least d neighbours. If $|D| > d$, every source neighbours D, since none can miss all its positions. In each case Hall's inequality holds. □

Lemma 6 (Missing pairs and shared corners)**.** *Every five-colour variety has eight distinct cyclic corner types. An adjacent-repeat variety lacks exactly one bichromatic edge type; an opposite-repeat variety lacks exactly two, which are disjoint. Two disjoint missing pairs determine a unique opposite-repeat variety. If either of two varieties is opposite-repeat, they share at least two corner types. A target has at most one monochromatic edge; each bichromatic type occurs on at most two edges, and a doubled type contains the repeated colour. Thus two disjoint bichromatic types account for at most three geometric target edges.*

Proof. A missing bichromatic pair occupies opposite singleton faces: if a colour is repeated, a face of the other colour cannot be opposite both copies. An adjacent-repeat variety has one opposite pair of singleton faces, and an opposite-repeat variety has two. Two disjoint missing pairs specify two opposite face pairs, leaving the fifth colour on the remaining pair, which determines a unique variety.

For the shared-corner assertion, let the opposite-repeat variety repeat r, with singleton opposite pairs $\{\alpha, \beta\}$ and $\{\gamma, \delta\}$. It contains both cyclic orientations of each rxy with $x \in \{\alpha, \beta\}$ and $y \in \{\gamma, \delta\}$. In the other variety, an r-face with no adjacent r-face has at least three distinct colours around it. Their cyclic order crosses between the two pairs at least twice, giving two shared corners. If its r-faces are adjacent, the other neighbours of each form a path of three distinct colours. Each path crosses between the pairs and gives a shared corner, and these have different types. Corner counts are symmetric: a five-colour variety has eight distinct cyclic corner types, since the only repeated colour multisets occur at the two ends of an edge, in opposite cyclic orders.

Only the repeated colour can form a monochromatic edge. Two singleton faces meet along at most one edge; hence a doubled bichromatic type contains the repeated colour and occurs at most twice. Two disjoint types cannot both contain that colour, proving the final bound. □

Lemma 7 (A transverse edge forces a common corner)**.** *Let source and target both be adjacent-repeat varieties. If the source lacks the type of a target edge not parallel to the target's monochromatic edge, then it shares a corner type with the target.*

Proof. Relabel the colours and rotate the target to $T = (0, 1, 0, 2, 3, 4)$. Its monochromatic edge is parallel to z. The types occurring on edges transverse to the monochromatic edge are $03, 04, 13, 14, 23, 24$. The proper half-turn interchanging the x- and y-directions and reversing z, followed by the colour relabelling $1 \leftrightarrow 2$, $3 \leftrightarrow 4$, fixes T and pairs these types as $(03, 04)$, $(13, 24)$, and $(14, 23)$. Thus three cases suffice. For each missing type the source has its two colours on opposite faces. The other three colours occur around that axis in one of two cyclic

orientations. The three adjacent-repeat choices in each orientation share six tricoloured corners. Reading those common corners gives:

Missing type	Shared corners in one orientation	Shared corners in the other orientation
03	$024, 041$	123
13	032	$041, 142$
14	$013, 123$	024

Every entry is nonempty, so the source and target share a corner. □

For a bichromatic pair $\alpha\beta$, let $\mathcal{H}_{\alpha\beta}$ be the family of varieties lacking that edge type. In every such cube, α and β are opposite singleton faces. The four faces around this axis use the other three colours, one of them twice. There are nine varieties in $\mathcal{H}_{\alpha\beta}$: three opposite-repeat varieties O_r, and six adjacent-repeat varieties A_r^+ and A_r^-, where r is the repeated colour. The signs distinguish the two cyclic orientations of the three distinct ring colours. All three A_r^+ varieties share the same six tricoloured corner types; all three A_r^- varieties share the reverse six types.

Lemma 8 (The missing-pair corner array)**.** *The eight corners of an opposite-repeat target $O_r \in \mathcal{H}_{\alpha\beta}$ form a 2×2 array with two positions per cell. Rows correspond to the two unordered ring-colour pairs involving r, and columns to their cyclic orientations. A source O_r fits every cell; the other two opposite-repeat varieties each fit one row; and the positive and negative adjacent-repeat varieties fit the corresponding columns. Consequently, every variety in $\mathcal{H}_{\alpha\beta}$ shares at least four corner types with every opposite-repeat variety in that family.*

Proof. Each cell contains one corner at each end of the $\alpha\beta$ axis. Reading the ring colours gives the stated incidences. Figure 1 shows the example $O_2 = (0, 1, 2, 2, 3, 4)$ in $\mathcal{H}_{01}$, with $+$ chosen to include 023. Relabelling the three ring colours gives the other two targets, and relabelling the palette gives every missing-pair family. Cyclic rotations identify corner labels. In particular, the two positions in each cell have identical source neighbourhoods. □

	A_r^+ sources positive column	A_r^- sources negative column
pair 23 O_3 fits this row	$023, 132$	$032, 123$
pair 24 O_4 fits this row	$042, 124$	$024, 142$

O_2 fits all four cells

Figure 1: Corner array of O_2 in the family lacking 01. Each cell contains two target positions. Here $r \in \{2, 3, 4\}$; all three adjacent-repeat sources of a given sign fit the same column.

3.1 Signed corner projections

Retain the family $\mathcal{H}_{\alpha\beta}$ from Lemma 8, with its opposite-repeat varieties and their 2×2 corner array. Write $C(W)$ for the cyclic corner types of W.

Lemma 9 (Corner projections onto a missing-pair family)**.** *Fix a bichromatic pair $\alpha\beta$. Let C^+ and C^- be the two sets of six tricoloured corner types shared by the three adjacent-repeat varieties of the respective cyclic orientations in $\mathcal{H}_{\alpha\beta}$. An opposite-repeat variety in $\mathcal{H}_{\alpha\beta}$ has four corners in each set. For a source $W \notin \mathcal{H}_{\alpha\beta}$, put*

$$p(W) = |C(W) \cap C^+|, \qquad m(W) = |C(W) \cap C^-|.$$

Then

$$(p(W), m(W)) \in \{(4,0),(2,1),(1,2),(2,2),(0,4)\}.$$

Proof. Relabel the poles as $0,1$ and the ring colours as $2,3,4$. Choose the positive orientation so that

$$C^+ = \{023, 034, 042, 124, 132, 143\}, \qquad C^- = \{024, 032, 043, 123, 134, 142\}.$$

Triples denote cyclic classes; face order is $(+x,-x,+y,-y,+z,-z)$. The following table gives representatives of all geometric cases outside $\mathcal{H}_{01}$, up to a colour permutation preserving the pole pair and a simultaneous reflection, which may exchange C^+ and C^-. Such a reflection is only a symmetry of this incidence calculation, not an allowed assembly move.

W	$p(W)$	$m(W)$
$(0,0,1,2,3,4)$	2	2
$(0,1,0,2,3,4)$	0	4
$(0,2,0,3,1,4)$	2	1
$(0,2,1,2,3,4)$	0	4
$(0,2,1,3,2,4)$	2	1
$(0,2,1,3,4,4)$	2	2

To see that these cases exhaust the possibilities, first suppose a pole colour is repeated. Its two faces are either opposite, or adjacent; in the latter case the other pole is opposite one repeated face or adjacent to both. If a ring colour is repeated, the two pole faces must be adjacent, since otherwise W belongs to $\mathcal{H}_{01}$. Fix them at x^+ and y^+. Among the remaining faces x^-, y^-, z^+, z^-, an opposite repeated pair is z^+, z^-. An adjacent repeated pair is either x^-, y^-, or one of x^-, y^- together with one of z^+, z^-. These are the final three rows.

Each entry follows by reading the eight corners of the displayed cube. The projections of the nine family varieties themselves are $(6,0),(0,6)$, and $(4,4)$, as follows directly from their ring descriptions. □

Lemma 10 (Compatibility for a thin frame)**.** *Choose an orientation of a five-colour target for a $2\times2\times3$ frame such that its four noncorner edge positions are bichromatic. Every source cube fits at least four of the twelve positions.*

Proof. If either source or target is opposite-repeat, Lemma 6 gives two shared corners. The four parallel bichromatic edges have either four distinct types or two types occurring twice each. In the latter case both contain the target's repeated colour. The source's missing pairs are disjoint, so it fits at least two of these edges, giving four compatible positions.

If both varieties are adjacent-repeat, the four parallel edge types are distinct and transverse to the monochromatic edge. The source fits either all four, or at least three and a corner by Lemma 7. Again it fits four positions. □

3.2 Eight family cubes and three arbitrary additions

Lemma 11 (Eight cubes with a common missing pair)**.** *Eight cubes from $\mathcal{H}_{\alpha\beta}$, together with any three additional five-colour cubes, have a corner solution whose target belongs to $\mathcal{H}_{\alpha\beta}$.*

Proof. Let O, P, Q count the opposite-repeat, positive adjacent-repeat, and negative adjacent-repeat copies among the chosen eight. Thus $O+P+Q=8$; C^+ and C^- are as in Lemma 9.

If $O+\min(P,Q)\geq 4$, the 2×2 Hall argument gives an opposite-repeat corner solution from these eight copies alone. Explicitly, choose O_r whose opposite-repeat multiplicity is maximal. Each cell has at least $O/2+\min(P,Q)\geq 2$ neighbours, each row has at least $8-O/2\geq 4$, and each column has at least $O+\min(P,Q)\geq 4$. Every other nonempty union of cells has all eight copies as neighbours.

Otherwise exchange the signs so that $O + Q \le 3$. Put $\kappa = O + Q = 8 - P$, so $P \ge 5$ and $0 \le \kappa \le 3$. Some positive variety A_r^+ occurs at least twice. If its eight corners have a matching among all eleven copies, we are done. Otherwise, choose a deficient Hall set D of its corners. The two copies of A_r^+ fit every corner, so $|D| \ge 3$ and D contains a tricoloured corner. All P positive copies are therefore neighbours of D. Let R be its other neighbours and put $U = C^+ \setminus D$, $u = |U|$. Since $|D| + u \le 8$,

$$P + |R| < |D| \le 8 - u, \qquad |R| + u \le \kappa - 1.$$

Remove the P positive copies and the copies in R. Every surviving copy has all its positive corners in U, and $u \le 2$.

Every family opposite-repeat variety has four corners in C^+, so no such chosen copy remains. At least

$$\kappa - |R| \ge u + 1$$

of the chosen nonpositive copies remain, and all of them are negative adjacent-repeat copies. Each fits every corner in C^-. The remaining pool has at least

$$\kappa + 3 - |R| \ge 4 + u$$

copies.

Lemma 9 gives the following implications for every source:

$\lvert C(W) \cap C^+ \rvert \le u$	$u = 0$	$u = 1$	$u = 2$
$\lvert C(W) \cap C^- \rvert \ge$	4	2	1.

Indeed, the family varieties have projection sizes $(6,0), (0,6), (4,4)$, and every variety outside the family has one of $(4,0), (2,1), (1,2), (2,2), (0,4)$.

Partition C^- into three pairs, indexed by the unordered pairs of ring colours. The negative half of each family opposite-repeat target is the union of two of these pairs. We will match one such four-position half using the remaining pool, then use four positive copies for its positive half.

Let j be the number of negative-family copies in the remaining pool; thus $j \ge u + 1$. If $j \ge 4$, the conclusion is immediate. If $j = 3$, another remaining copy fits at least one negative corner; choose a target whose negative half contains it and fill the other three positions with negative-family copies.

If $j = 2$, then $u \le 1$. Two other remaining copies each fit at least two negative corners, so they can be assigned distinct ones. Any two negative corners lie in the union of two of the three pairs. Choose the corresponding opposite-repeat target and fill its other two negative positions with the two negative-family copies.

Finally, if $j = 1$, then $u = 0$, and there are at least three other copies, each fitting at least four of the six negative corners. Choose one of these copies W. Some union of two of the three pairs contains at least three of its negative corners. In that four-position half, every other copy fits at least two positions. Hence W and any two other copies can be matched to three distinct positions: two copies have at least two neighbours, and all three together have at least three. The one negative-family copy fills the fourth position.

In every case we obtain the negative half of an opposite-repeat target in $\mathcal{H}_{\alpha\beta}$. Four of the $P \ge 5$ positive copies fill its positive half. The positive copies used here are disjoint from the surviving pool. This proves the lemma. □

4 An elementary fourteen-cube obstruction

Consider four copies of A, five of B, and five of C, where

$$A = (0,0,1,2,3,4), \quad B = (0,1,2,3,2,4), \quad C = (0,3,1,2,4,1).$$

Their corner types are shown below. Triples denote cyclic classes; the last column counts positions whose type occurs in neither of the other varieties.

Variety	Corner types	Private positions
A	$013, 014, 023, 024, 031, 032, 041, 042$	5
B	$022, 023, 034, 042, 122, 124, 132, 143$	6
C	$011, 014, 021, 042, 113, 123, 134, 243$	6

Lemma 12. *The collection $4A + 5B + 5C$ has no five-colour solution.*

Proof. A solvable target must have every corner type in the displayed union. Its only two-colour types are $022, 122, 011, 113$. An adjacent-repeat target therefore repeats either colour 2, with types $022, 122$, or colour 1, with types $011, 113$. In either case these two corners determine the colours on the faces at the ends of the repeated-colour edge. The two orders of the remaining colours give a mirror pair: B and its mirror, or C and its mirror. The mirror of B requires 043, and the mirror of C requires 012; both types are absent. Only B, C remain.

An opposite-repeat target requires both cyclic orientations of each corner triple. The unordered triples with both orientations in the union are

$$013, 014, 023, 024, 123, 134.$$

If its repeated colour is r, delete r from the four triples containing it. The remaining pairs must form a $K_{2,2}$ on the other four colours: choose one face from each of their two opposite pairs. Colours 2 and 4 occur in fewer than four listed triples. For $r = 1$ or $r = 3$, the four pairs have degree sequence $(3, 2, 2, 1)$, so do not form $K_{2,2}$. For $r = 0$, the pairs are $13, 14, 23, 24$, forcing opposite pairs $\{1, 2\}$ and $\{3, 4\}$. This is A.

Thus only A, B, C can be targets. Their private positions require respectively at least five, six and six copies of the corresponding variety, exceeding the available four, five and five copies (Figure 2). None is solvable. □

5 An elementary upper bound for corners

We prove that every fifteen sources have a corner assembly. The argument uses the two kinds of repeated colour differently: opposite-repeat cubes supply both orientations of certain corners, whereas adjacent-repeat cubes supply one orientation and two corners involving their repeated colour. The longer geometric reductions are proved in the separately paginated *Mathematical appendix to Sharp thresholds for five-colour cube assemblies*. That appendix is part of the mathematical argument, intended for refereeing with this article; the checking programs are supplementary.

5.1 Directed corners and repeated colours

Encode an oriented tricoloured corner (i, j, k) by the complementary ordered pair ab, chosen so that (a, b, i, j, k) is an even permutation of the five colours. Cyclic permutation of (i, j, k) does not change ab; reversing it replaces ab by ba. Write $V = \{0, 1, 2, 3, 4\}$. The six tricoloured corners of an adjacent-repeat variety form a directed cut

$$C(S) = \{ab : a \in S,\ b \in V \setminus S\}, \qquad |S| \in \{2, 3\}.$$

The smaller side P of the cut is its *pole pair*. For each cut there are three varieties, indexed by their repeated colour $r \notin P$. Their other two corners are rrp, for the two $p \in P$. Thus there are twenty adjacent classes, each consisting of three varieties. Class multiplicity always counts

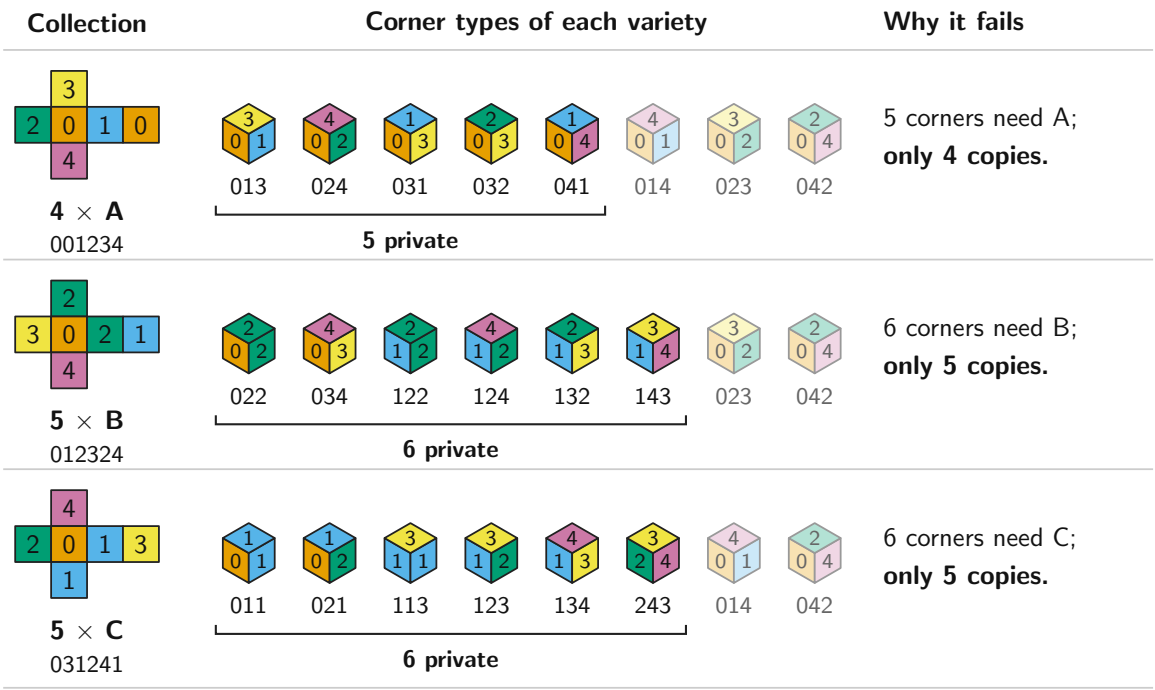


Figure 2: The fourteen-cube obstruction $4A + 5B + 5C$. Each net has $+x$ at its centre. Corner types are read counterclockwise from outside; the full-colour types occur on no other variety. A target of variety X has these private positions, and only copies of X fit them.

physical copies, allowing different repeated colours within a class. An opposite-repeat variety is represented by a four-cycle on four vertices of V: it supplies both directed corners on each edge, and the omitted vertex is its repeated colour. Equivalently its edges form a $K_{2,2}$. These descriptions follow by reading the corners around the repeated faces.

For a class with at least two copies, call a six-corner matching after reserving two of its copies an *own completion*. The reserved copies need not have the same repeat, so this alone is not an assembly.

Lemma 13 (Repeated-colour completion)**.** *An adjacent class with an own completion assembles its target if two of its copies have the same repeated colour. If two classes each have three copies and an own completion, and their pole pairs intersect, the pool has an assembly.*

Proof. A same-repeat pair fills the two bicoloured corners, leaving precisely the six-corner matching. Otherwise each triple must use all three repeats allowed by its pole pair. The two repeat sets intersect; choose a common colour r. Reserve one copy from each class, using the second class at the shared pole of the first target. Start with the first class's own completion. Restore one of its two reserved copies and remove the chosen copy of the second class. If the latter was used, the restored copy, which fits every tricoloured corner of this target, replaces it. The remaining six positions are still matched to distinct sources. □

Lemma 14 (A nine-source construction)**.** *For distinct colours a, b, c, d, e, an opposite source containing the edge ab, together with three sources in each of $C(ac), C(bc)$ and two in $C(abc)$, has an assembly. The same holds after reversing all cuts.*

Proof. Put $A = C(ac)$, $B = C(bc)$ and $H = C(abc)$. After reserving two A copies, match its six tricoloured corners by

$$A : cb, \qquad B : cd, ce, \qquad H : ad, ae, \qquad O : ab.$$

After reserving two B copies use

$$B : ca, \qquad A : cd, ce, \qquad H : bd, be, \qquad O : ba.$$

A class assigned two positions supplies two distinct copies. Both triples have own completions, and their poles meet at c, so Lemma 13 applies. □

5.2 Structural inputs

The appendix proves the following concentration and assembly statements. Their proofs use the cut model, Hall's theorem, repeated-colour completion, and explicit geometric cases. None assumes the corner threshold. The prescribed-family lemma 11, proved earlier, is used in two concentrated-opposite cases.

Lemma 15 (Concentration)**.** *A failing fourteen-source pool containing at least three copies of an opposite variety has one of these two profiles: four copies of that variety and adjacent-class sizes* $5, 5$*, or three copies and adjacent-class sizes* $5, 3, 3$*. In either case there are no other opposite sources. A mixed fourteen-source pool in which every opposite variety has multiplicity at most two and some adjacent class has multiplicity at least four has an assembly.*

The opposite profiles are proved in the appendix arguments headed "Four identical opposite copies" and "Three identical opposite copies"; the mixed concentration arguments then treat the possible opposite totals. The conclusion records class sizes, not repeated-colour refinements.

Lemma 16 (Assembly inputs)**.** *The following pools have assemblies:*

1. *fifteen adjacent sources;*
2. *one opposite source and fourteen adjacent sources;*
3. *three opposite sources and eleven adjacent sources, with at most three copies in each adjacent class;*
4. *five opposite sources, each variety occurring at most twice, and sources* $2A + B + C$ *in three distinct adjacent classes;*
5. *nine adjacent sources in distinct classes;*
6. *eight opposite sources, each variety occurring at most twice;*
7. *seven opposite sources with that same cap and any adjacent source.*

The appendix gives complete proofs, including the geometric exhaustions behind the finite Hall tables. Item 3 includes all three opposite multiplicity patterns, including three identical varieties. Items 4–7 produce opposite targets; the other items may produce either kind of target. These distinctions will matter when a source is replaced.

For the present consolidation draft, the principal appendix locations are:

Input	Appendix argument
Concentrated opposite profiles	22, 14
Fifteen adjacent sources	39
One opposite and fourteen adjacent	52
Three opposite and eleven adjacent	66
Five-opposite helper construction	30
Nine distinct adjacent classes	29
Eight opposite; seven plus one adjacent	10
Butterfly; two stars; star and triangle	54, 63, 59

These locations include references to the supporting geometric arguments; they do not replace those arguments.

For orientation, the proof of item 1 is a deletion argument. A failing pure fourteen-pool has class cap three, at most eight occupied classes, and at most five repeated classes. If none of its triples has an own completion, its class sizes are $3, 3, 3, 3, 1, 1$. A failing pure fifteen-pool with class cap three cannot have two own-completed triples. Deleting a copy of any triple leaves a doubled class, hence forces an own-completed triple elsewhere. Restore the copy and delete from this new triple: a second own-completed triple is forced, a contradiction. For item 2, the appendix proves that a pure fourteen-failure has at least two triples. A triple without an own completion has a private column in its two-by-three corner array. The added opposite source either meets that column, creating a second own completion, or belongs to one of five avoiding varieties, whose three geometric types have direct completions. These explanations are reading guides; the appendix supplies the required concentration, private-column and repeated-colour proofs.

5.3 Compressing two adjacent sources

Lemma 17 (Disjoint-cut compression)**.** *Two disjoint adjacent cuts have an opposite variety's eight corners in their union. Replacing two actual sources from these classes by one virtual source of that opposite variety preserves the implication "the smaller pool assembles ⇒ the original pool assembles".*

Proof. Normalize the first cut to $C(01)$. The disjoint cuts are exactly $C(234), C(23), C(24), C(34)$. For $C(ij)$ take the opposite variety with bipartition $01 \mid ij$; for $C(234)$ choose any two of $2, 3, 4$. The first actual source supplies one directed half and the second the other. An assembly assigns only one position to the virtual source. Replace it by whichever removed source supplies that corner. Both removed copies were unavailable to all other positions, so the eight physical sources remain distinct. A virtual opposite source can only have been assigned a tricoloured corner, even if the target is adjacent. □

Lemma 18 (Intersecting cuts)**.** *A pairwise-intersecting family of adjacent cuts is contained, up to colour permutation and simultaneous reversal, in one of the following families:*

Butterfly	$C(01), C(02), C(12), C(123), C(124), C(012)$
Two stars	$C(01), C(02), C(03), C(04), C(034), C(024), C(023)$
Star and triangle	$C(01), C(02), C(03), C(04), C(034), C(024), C(014)$.

Proof. Represent a size-two outset by its positive pole pair and a size-three outset by its negative pole pair. Same-sign cuts are disjoint exactly when their pairs are disjoint; opposite-sign cuts are disjoint exactly when their pairs agree. The two signed edge families in K_5 are therefore intersecting and share no edge. Each intersecting edge family lies in a star or triangle: after edges ab, ac, an edge not containing a must be bc. Two complete triangles without a common edge meet in one vertex, giving the butterfly. For a triangle and a star, a centre outside the triangle gives the star-and-triangle family. A centre inside permits only its two edges to the outside vertices; extend them to a second triangle. For two stars with different centres, assign their common edge to the sign already using it, if either, then extend both families to obtain the two-star form. With the same centre and at least two edges in each family, each has exactly two edges; complete each to a triangle. If one family has at most one edge, choose its other endpoint as centre, or any different centre if it is empty, reducing to the different-centre case. □

The appendix proves that thirteen adjacent sources of class cap three in any one of these families assemble after any two opposite sources are adjoined. These are the butterfly, two-star and star-and-triangle assembly arguments. The last reduces, by two small matching lemmas, to the nine-source construction in Lemma 14.

Proposition 19. *Two opposite sources and thirteen adjacent sources of class cap three have an assembly.*

Proof. If a disjoint pair of adjacent classes occurs, compress one copy of each. The resulting pool has three opposite and eleven adjacent sources; Lemma 16(3) applies, and Lemma 17 lifts the assembly. Otherwise Lemma 18 and the three family arguments apply. □

5.4 The fifteen-source conclusion

Proof of Theorem 3. Lemma 12 proves the lower bound. Suppose that a fifteen-source pool fails. Any subpool fails. Five copies of one opposite variety are excluded by Lemma 15, retaining them in a fourteen-subpool. If an opposite variety occurs three or four times, the same lemma excludes any other opposite variety.

Put $A = \sum_j \binom{g_j}{2}$, where g_j are adjacent-class counts. With four opposite copies, deleting any of the eleven adjacent sources leaves class sizes $5, 5$ and pair count 20. Each original same-class pair survives nine deletions, giving $9A = 11 \cdot 20 = 220$, impossible. With three opposite copies the twelve deletions instead give $10A = 12(\binom{5}{2} + 2\binom{3}{2}) = 192$, also impossible. Every opposite variety therefore occurs at most twice. In a mixed pool, four copies of an adjacent class could be retained with an opposite source in a fourteen-subpool, contradicting the concentration lemma. Thus mixed pools also have adjacent-class cap three.

Let o be the number of opposite sources. For $o = 0, 1$, use Lemma 16(1),(2). For $o = 2$, use Proposition 19. For $o = 3, 4$, retain three opposite and eleven adjacent sources and use item 3. For $o = 5, 6$, retain five opposite and nine adjacent sources. If the latter occupy nine classes, item 5 applies. Otherwise some class A occurs at least twice; the cap of three ensures at least three occupied classes, so select $2A + B + C$ and use item 4. For $o = 7$, item 7 applies, and for $o \geq 8$ retain eight opposite sources and use item 6. Each possibility yields an assembly, contradicting failure. Hence $N_5 = 15$. □

6 Rectangular boxes

We prove Theorem 1. The shared corner lemmas and Theorem 3 are now available. A uniform lower obstruction is followed by the thin-box upper bounds and the remaining shapes. These deductions use no computational premise.

After ordering the sides as $a \leq b \leq c$, the upper bound has three exhaustive regimes:

Shape	Main mechanism
$a + b \geq 6$	Weighted compatibility and reservations
$a = b = 2$	Concentration and retargeting along a rod
$a = 2,\ b = 3$	Exceptional-defect control and retargeting

We establish the thin regimes first, then the general weighted bound. In each case an initial corner matching verifies part of Hall's inequalities; the final matching may rearrange those placements.

Lemma 20 (Rare pairs)**.** *In a collection of five-colour cubes, any family of bichromatic pairs each missing from more than half the source copies consists of at most two pairwise disjoint pairs.*

Proof. The missing families of two such pairs intersect. A source missing both pairs forces them to be disjoint, by Lemma 6. There are at most two disjoint pairs in a five-colour palette. □

6.1 A uniform lower bound

Lemma 21 (A three-variety obstruction)**.** *Let $a, b, c \geq 2$ and put $\tau = a + b + c - 6$. In face order $(+x, -x, +y, -y, +z, -z)$, define*

$$U = (0, 1, 0, 2, 3, 4), \qquad V = (0, 1, 1, 2, 3, 4), \qquad W = (0, 3, 1, 4, 2, 3).$$

The collection

$$U + (\tau + 5)V + (\tau + 6)W$$

of $2\tau + 12$ cubes has no $a \times b \times c$ frame using all five colours. In particular, $U + 7V + 8W$ fails for both a $2 \times 3 \times 3$ frame and a $2 \times 2 \times 4$ box.

Proof. The corner types are

U	$003, 004, 013, 024, 032, 041, 123, 142$
V	$113, 114, 013, 024, 032, 041, 123, 142$
W	$133, 334, 012, 024, 031, 043, 132, 234.$

Triples denote cyclic classes. The only two-colour types in their union are $003, 004$, $113, 114$, and $133, 334$. An adjacent-repeat target must therefore repeat colour 0, 1, or 3. Its two repeated-colour corners force the colours on the opposite faces at the ends of its monochromatic edge to be, respectively, $\{3, 4\}$, $\{3, 4\}$, or $\{1, 4\}$. The two orders of the remaining colours give U, V, W, respectively, and their mirrors. The mirrors of U and V require 023, which is absent from the union; the mirror of W requires 021, also absent.

An opposite-repeat target requires both cyclic orientations of four distinct unordered tri-coloured triples. The union above contains both orientations only for 013 and 123. Thus no opposite-repeat target is possible. The only corner-supported targets are U, V, W.

Target U needs two copies possessing a 00 corner, but the collection has only one such copy.

For target V, seven of its corner types are absent from W; the only shared type is 024. In addition, exactly one geometric edge in each coordinate direction has a type absent from W: these are $14, 14, 11$. Consequently, in any orientation of an $a \times b \times c$ frame, $7 + \tau$ positions require a U or V source. There are only $\tau + 6$ such copies.

For target W, seven corner types are absent from both U and V. Again, one geometric edge in each direction has a private type, now $34, 33, 34$. Hence $7 + \tau$ positions require a W source, whereas only $\tau + 6$ are available.

Every possible target therefore fails (Figure 3). □

Proposition 22. *For every $a, b, c \geq 2$,*

$$F_5(a, b, c) \geq \max\{15,\ 4(a + b + c) - 16,\ 2(a + b + c) + 1\}.$$

Proof. The three terms follow from the fourteen-cube corner obstruction (Lemma 12), the number of frame positions, and Lemma 21, respectively. □

6.2 The smallest thin box

Theorem 23 (The $2 \times 2 \times 3$ box)**.** *We have*

$$F_5(2, 2, 3) = 15.$$

In particular, any fifteen five-colour cubes contain twelve that form a $2 \times 2 \times 3$ box with monochromatic exterior faces using all five colours.

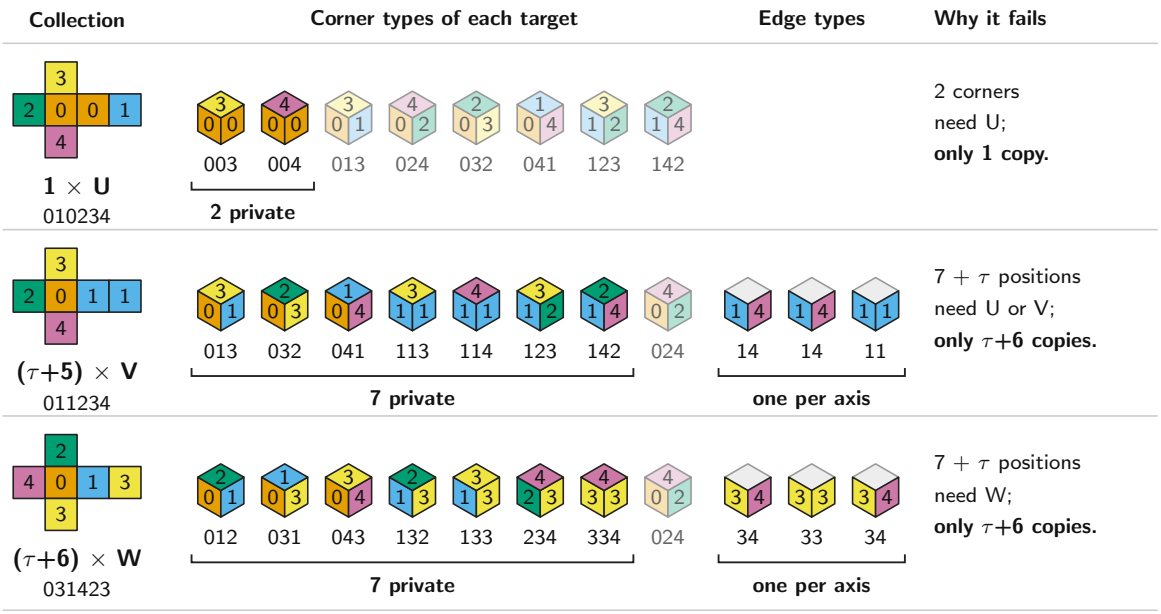


Figure 3: The uniform obstruction $U + (\tau + 5)V + (\tau + 6)W$, where $\tau = a + b + c - 6$. Each net has $+x$ at its centre, and corner types are read counterclockwise from outside. Only U, V and W can be targets. Their pools are U alone, U or V, and W alone; faded corner types also occur on a cube outside the pool. For V and W, one edge in each direction has a private type, and these edges have τ noncorner positions in total.

Proof. The fourteen-cube corner obstruction (Lemma 12) gives the lower bound: the eight corner cubes of any box would themselves give a corner solution.

For the upper bound, take fifteen arbitrary source copies. Call a bichromatic pair rare if at most seven sources possess it. Each rare pair is missing from at least eight sources. The missing sets for any two rare pairs intersect, since $8 + 8 > 15$. Lemma 20 therefore gives at most two disjoint rare pairs.

Suppose first that there is no rare pair. The corner theorem supplies a target T and a corner matching. If T is adjacent-repeat, choose the long axis perpendicular to its monochromatic edge. If T is opposite-repeat, any choice of long axis is suitable. In either case all four noncorner frame positions have bichromatic requirements, and each requirement has at least eight source neighbours.

If there is a rare pair $e = \alpha\beta$, at least eight sources belong to $\mathcal{H}_{\alpha\beta}$. Choose eight of them and any three additional copies. Lemma 11 gives a corner matching with target $T \in \mathcal{H}_{\alpha\beta}$. Orient its $\alpha\beta$ axis along one of the two short directions, and choose a perpendicular direction as the long axis. The four noncorner edge requirements then each join a pole colour, α or β, to a ring colour. None is $\alpha\beta$, and none can be the other rare pair, if present, since that pair is disjoint from $\alpha\beta$ and uses only ring colours. The target's monochromatic edge, when present, is parallel to the $\alpha\beta$ axis and contributes no noncorner position. Again all four edge requirements are bichromatic and have at least eight source neighbours.

For either chosen target and orientation, every source fits at least four of the twelve positions, by Lemma 10. Thus each source misses at most eight positions.

Let D be any subset of the twelve target positions. If D contains only corners, its Hall inequality follows from the corner matching. Otherwise, if $|D| \leq 8$, one of its edge positions has at least eight neighbours. If $|D| > 8$, every source neighbours D, because no source misses more than eight positions. Hall's theorem supplies a matching of all twelve positions to distinct source copies. The final matching may rearrange the preliminary corner placements. □

6.3 Sharp thresholds for $2 \times 3 \times c$ frames

Lemma 24 (Compatibility for $2\times3\times c$)**.** *Let $M = c-2 \geq 2$ and put $d = 2M+6$. Orient a five-colour target for a $2\times3\times c$ frame so that its monochromatic edge, if any, is parallel to the side of length two. An adjacent-repeat source misses at most d positions, and an opposite-repeat source misses at most $d+1$ positions. If a source misses $d+1$ positions, it is the unique opposite-repeat variety lacking the two bichromatic types appearing among those missed edge positions.*

Proof. The nonzero edge lengths are $1, M$. By Lemma 6, for an opposite-repeat source, at most six corners are incompatible. Its two missing bichromatic pairs are disjoint, so at most three geometric edges are incompatible, with at most two parallel to either nonzero direction. If their counts are k_1, k_M, the defect is at most

$$6 + k_1 + Mk_M \leq 6 + 3 + 2(M-1) = 2M + 7 = d + 1.$$

Equality requires six incompatible corners, two incompatible long edges, and one incompatible unit-length edge. These three geometric edges display exactly two bichromatic types: there are at least two because any target type occurs at most twice, and at most two because the source lacks only two types. They are disjoint and determine the opposite-repeat source variety uniquely.

An adjacent-repeat source lacks only one bichromatic type. Against an opposite-repeat target, at most six corners and edge positions of total weight $2M$ are incompatible, giving defect at most d. Against an adjacent-repeat target, an incompatible edge lies transverse to the monochromatic edge, so Lemma 7 gives a shared corner. The absent type contributes weight at most $M+1$: a doubled type has one edge in each nonzero direction. The defect is therefore at most

$$7 + (M+1) = M + 8 \leq 2M + 6 = d.$$

If no actual edge is incompatible, the defect is at most eight, which is also at most d. □

Lemma 25 (An opposite-repeat target along the short axis)**.** *Let $c \geq 3$. For a $2 \times 3 \times c$ frame, orient an opposite-repeat target with its repeated-colour axis along the side of length two. Every source misses at most $2(c-2)+6$ positions.*

Proof. Every actual edge type contains the target's repeated colour. By Lemma 6, a source can lack at most one such bichromatic type, because its missing pairs are disjoint. The absent type accounts for at most two parallel geometric edges, of total weight at most $2(c-2)$. At least two corners are shared, so at most six further positions are missed. □

Theorem 26 (The remaining $2 \times 3 \times c$ frames)**.**

$$F_5(2,3,3) = 17, \qquad F_5(2,3,c) = 4c+4 \quad (c \geq 4).$$

Proof. First let $c = 3$ and take seventeen sources. Call a bichromatic pair rare if at most eight sources possess it. Each rare pair is absent from at least nine sources. By Lemma 20 there are at most two, and they are disjoint. If no pair is rare, choose a corner target using $N_5 = 15$ and put its possible monochromatic edge along the side of length two. Otherwise choose a rare pair $\alpha\beta$ and apply Lemma 11 to eight sources in $\mathcal{H}_{\alpha\beta}$ and any three others. Orient the target's $\alpha\beta$ axis along the side of length two. All actual edge types join a pole colour to a ring colour, so they avoid both possible rare pairs and are bichromatic.

Every source misses at most nine of the sixteen frame positions. Indeed, if source or target is opposite-repeat, at most six corners and three actual geometric edges are incompatible. If both are adjacent-repeat, at most two actual geometric edges are incompatible; whenever one is incompatible, Lemma 7 supplies a shared corner. If no edge is incompatible, at most eight positions are missed.

The chosen target has a corner matching, and every edge requirement has at least nine neighbours. Lemma 5, with the corners as the matched subset, completes the frame. This proves $F_5(2,3,3) \le 17$; the collection $U + 7V + 8W$ in Lemma 21 gives equality.

Now let $c \ge 4$, put $M = c - 2 \ge 2$, and set

$$d = 2M + 6, \qquad N = 4M + 12 = 2d.$$

There are $N = 4c + 4$ frame positions. Take any N source copies, and call a bichromatic pair rare if at most $d-1$ copies possess it. Each rare pair is absent from at least $d+1 > N/2$ sources. Lemma 20 gives at most two disjoint rare pairs.

If no pair is rare, use the corner theorem and orient the target's possible monochromatic edge along the side of length two. Otherwise, choose a rare pair $\alpha\beta$. At least $d+1 \ge 11$ sources belong to $\mathcal{H}_{\alpha\beta}$. Lemma 11 supplies a corner solution whose target belongs to that family. Orient its $\alpha\beta$ axis along the side of length two. All actual edge types join a pole colour to a ring colour, and hence avoid both rare pairs; the monochromatic edge, if present, has no noncorner positions.

In either case the chosen target has a corner matching and every actual edge requirement has at least d source neighbours. Hall's inequality holds for corner-only sets, and for any set of at most d positions containing an edge. By Lemma 24, sets of more than $d+1$ positions neighbour every source.

Suppose the frame has no matching. A deficient Hall set D must therefore have exactly $d+1$ positions. Every nonneighbour misses all of those positions, so Lemma 24 says that all nonneighbours have one opposite-repeat variety V, determined by the two absent bichromatic types. Since D has at most d neighbours among $N = 2d$ copies, the inventory contains at least d copies of V.

Retarget the frame to V with its repeated-colour axis along the side of length two. By Lemma 25, every source misses at most d positions. The d copies of V are compatible with every position, so they verify Hall's inequality for every set of size at most d. Larger sets neighbour every source. This gives a frame after all.

Thus N sources always suffice. The number of frame positions supplies the matching lower bound. □

6.4 Sharp thresholds for $2 \times 2 \times c$ boxes

Put $M = c-2$. The frame of a $2\times2\times c$ box consists of eight corners and four parallel geometric edges, each with M positions. Its size is $L = 4M + 8 = 4c$. We call an orientation of an adjacent-repeat target *transverse* if its monochromatic edge is not parallel to the long axis. For an opposite-repeat target, its *axial* orientation has the repeated faces perpendicular to the long axis.

Lemma 27 (Directional compatibility)**.** *Let $M \ge 2$.*

1. *For a transverse adjacent-repeat target, every adjacent-repeat source misses at most $M+7$ positions. Every opposite-repeat source misses at most $2M + 6$ positions. An opposite-repeat source missing more than $M+6$ positions must be one of two varieties; their missing long-edge types are complementary pairs.*

2. *For an axial opposite-repeat target V, every source misses at most $d = 2M + 4 = L/2$ positions. If $M \ge 3$, a source different from V can miss d positions only if it is one of the other two opposite-repeat varieties with the same repeated colour. These two varieties have complementary missing long-edge types.*

Proof. For (i), write the target in face order $(+x, -x, +y, -y, +z, -z)$ as

$$(r, \alpha, r, \beta, \gamma, \delta),$$

and take the long direction to be x. Its four long-edge types are

$$r\gamma, \quad r\delta, \quad \beta\gamma, \quad \beta\delta.$$

An adjacent-repeat source lacks at most one of these. If it lacks one, Lemma 7 gives a shared corner, so its defect is at most $M+7$. If it lacks none, its defect is at most eight, also bounded by $M+7$.

By Lemma 6, an opposite-repeat source shares at least two corners and lacks at most two long-edge types, giving defect at most $2M+6$. To lack two of the four displayed types, its two missing pairs must be one of the disjoint matchings

$$\{r\gamma, \beta\delta\} \qquad \text{or} \qquad \{r\delta, \beta\gamma\}.$$

Each matching determines exactly one opposite-repeat variety, with repeated colour α. The two varieties have complementary missing long-edge types. Every other opposite-repeat source lacks at most one long-edge type, and hence has defect at most $M+6$.

For (ii), let the repeated colour of V be r, and put its singleton opposite pairs on the two transverse axes. The four long-edge types are the four edges of the resulting $K_{2,2}$ on the singleton colours. The other two opposite-repeat varieties with repeated colour r each share four corners and two long-edge types with V. Their compatible position count is $4+2M=d$, and their missing long-edge pairs are complementary.

Every adjacent-repeat source lacks at most one long-edge type. An opposite-repeat source with repeated colour different from r also lacks at most one: one of its two missing pairs contains r, whereas no long-edge type does. By Lemma 6, these sources share at least two corners with V, so they fit at least $2+3M \geq d$ positions. If $M \geq 3$, this last inequality is strict. The target variety V itself fits every position. This proves both assertions. □

Lemma 28 (Concentrated opposite-repeat copies)**.** *Let $M \geq 3$ and $d = 2M+4$. A collection of $2d = 4M+8$ cubes containing at least $d-1$ copies of one opposite-repeat variety forms a $2 \times 2 \times (M+2)$ box.*

Proof. Let that variety be V with multiplicity v. Use its axial target orientation. Every source misses at most d positions. If $v \geq d$, Hall's inequality is immediate: sets of at most d positions have the v copies of V as neighbours, and larger sets neighbour every source.

Suppose $v = d-1$. The same reasoning handles all position sets except sets D of size exactly d. If such a set fails Hall, no non-V copy neighbours it. Those copies must each miss d positions, so Lemma 27(ii) restricts them to the other two opposite-repeat varieties with the same repeated colour. Since $d > 8$, D contains a long-edge position. The two candidate varieties have complementary missing long-edge types, so at most one can be absent from the neighbourhood of D. All $2d - v = d+1$ non-V copies are therefore one variety W. Using W as the axial target gives the previous case, with at least d universal copies, and completes the box. □

Lemma 29 (A target avoiding scarce long-edge types)**.** *Suppose a collection has $N \geq 15$ cubes. Call a bichromatic pair rare when at most ρ sources possess it, where $N-\rho \geq 8$ and $2(N-\rho) > N$. Then there is a corner-solvable target, transverse if adjacent-repeat and axial if opposite-repeat, whose four long-edge types are all non-rare.*

Proof. Each rare pair is missing from more than half the collection. Thus any two rare pairs are disjoint and there are at most two, by Lemma 20. If there is no rare pair, use the corner theorem and choose either specified orientation.

Otherwise let $\alpha\beta$ be rare. At least eight sources belong to $\mathcal{H}_{\alpha\beta}$. Lemma 11 supplies a corner target in this family. Its $\alpha\beta$ axis is a singleton opposite pair and can be put along a short direction. If the target is opposite-repeat, choose its repeated-colour axis as the long direction; otherwise choose either axis perpendicular to the $\alpha\beta$ axis. Every long-edge type then joins one of α, β to a ring colour. It is neither $\alpha\beta$ nor the other possible rare pair, whose colours are both ring colours. These orientations are axial or transverse as required. □

Theorem 30 (Thin rods)**.**

$$F_5(2,2,4) \le 17, \qquad F_5(2,2,c) = 4c \quad (c \ge 5).$$

Proof. First let $c = 4$, so $M = 2$ and $L = 16$, and take seventeen source copies. In Lemma 29 use $\rho = 8$. The chosen target has a corner matching, and every long-edge type has at least nine sources. If the target is opposite-repeat, its defect is at most eight by Lemma 27, and Hall's theorem completes it.

For an adjacent-repeat target, adjacent-repeat sources miss at most nine positions and opposite-repeat sources at most ten. A failing Hall set D cannot consist only of corners. Since it contains an edge position, it has at least nine neighbours; and if $|D| > 10$, it neighbours every source. Therefore a failure would have $|D| = 10$ and exactly nine neighbours. Its eight non-neighbouring source copies must all be opposite-repeat. Each must miss two long-edge types, since otherwise its defect is at most $M + 6 = 8$. The two candidate varieties in Lemma 27(i) have complementary missing long-edge types. As D contains an edge position, all eight copies must be the same opposite-repeat variety V.

The axial target V is solved by Hall directly: its eight copies neighbour every nonempty set of at most eight positions, and larger sets neighbour all seventeen sources, since each misses at most eight positions. This proves the seventeen-cube upper bound.

Now let $M = c-2 \ge 3$, put $d = 2M+4$, and take exactly $N = 2d = L$ cubes. Use $\rho = d-1$ in Lemma 29. Every long-edge type of the chosen target has at least d sources. An opposite-repeat target is completed directly by Hall, since every source misses at most d positions.

For an adjacent-repeat target, adjacent-repeat sources miss at most $M + 7 \le d$ positions; opposite-repeat sources miss at most $2M + 6 = d + 2$. Any failing Hall set D therefore has

$$d + 1 \le |D| \le d + 2.$$

It has at most $d + 1$ neighbours, leaving at least $d - 1$ non-neighbouring source copies. All are opposite-repeat. An opposite-repeat source missing only one long-edge type has defect at most $M + 6 < d + 1$, so all these copies are among the two varieties in Lemma 27(i). Since D contains an edge, all must be one variety V. Lemma 28 then supplies a box from the original collection. Thus every collection of $L = 4c$ cubes works. The number of positions gives the matching lower bound. □

Proposition 31 (Complete rod threshold)**.** *Together with Theorem 3, Theorem 23 and the sixteen-cube obstruction of Lemma 21, Theorem 30 gives*

$$F_5(2,2,c) = \max\{15,\ 4c,\ 2c+9\} \qquad (c \ge 2).$$

Equivalently, the values are $15, 15, 17$ *for* $c = 2, 3, 4$*, respectively, and* $4c$ *for every* $c \ge 5$*.*

Proof. The cases $c = 2, 3$ are the established corner and smallest thin-box thresholds. For $c = 4$, Lemma 21 and the seventeen-cube upper bound give equality. For $c \ge 5$, the number of positions gives $F_5(2,2,c) = 4c$, as proved above. □

6.5 Boxes whose two shortest sides sum to at least six

Order the side lengths as $2 \le a \le b \le c$ and put

$$m = a - 2, \qquad t = b - 2, \qquad M = c - 2, \qquad \tau = m + t + M.$$

The frame has $L = 8 + 4\tau$ positions. In this section assume $m + t \ge 2$.

Lemma 32 (Weighted compatibility)**.** *Let* $2 \le a \le b \le c$ *with* $a + b \ge 6$*, so that* $m + t \ge 2$ *and* $M \ge 1$*. Every source misses at most* $d = \tau + M + 6$ *positions of every target frame.*

Proof. By Lemma 6, if the target is opposite-repeat, at most three geometric edges are incompatible, with at most two in any direction. Their total weight is at most $2M + t \le \tau + M$. At most six corners are incompatible, so the claimed bound follows.

For an opposite-repeat source, take the adjacent-repeat target

$$(r, \alpha, r, \beta, \gamma, \delta),$$

whose monochromatic edge is parallel to z. A doubled bichromatic type, $r\gamma$ or $r\delta$, occupies one edge in each of the x, y directions. If such a type is missing, it and the monochromatic edge contribute one edge in each direction; the other disjoint missing type contributes at most one further edge. The missed weight is at most $\tau + M$. If neither missing type is doubled, two missing single types cannot have the same direction: the single types in each direction share a colour. Adding the monochromatic edge therefore gives at most three edges, with at most two parallel, again of weight at most $2M + t \le \tau + M$. There are at most six incompatible corners.

For two adjacent-repeat varieties, at most three geometric edges are incompatible. If there are at most two, the total defect is at most

$$2M + 8 \le \tau + M + 6,$$

because $m + t \ge 2$. If there are three, they are the monochromatic edge and a doubled missing bichromatic type, one edge in each direction. Lemma 7 gives a shared corner. The defect is at most $\tau + 7 \le \tau + M + 6$, since $M \ge 1$. □

Theorem 33. *If $2 \le a \le b \le c$ and $a + b \ge 6$, then*

$$F_5(a, b, c) = 4(a + b + c) - 16.$$

Proof. Take $N = L = 8 + 4\tau$ source copies and put $d = \tau + M + 6$. Call a bichromatic pair rare if at most $d - 1$ copies possess it. Its missing family has at least

$$N - d + 1 = 3\tau - M + 3$$

copies, exceeding $N/2$ by $m + t - 1 \ge 1$. Lemma 20 again gives at most two disjoint rare pairs.

By Lemma 5, a target avoiding all rare pairs has a full frame matching whenever its corners and monochromatic edge positions have a matching: every other position has at least d neighbours, and Lemma 32 bounds each source's defect by d.

No rare pair. For each colour reserve up to m sources having that colour on adjacent faces, taking all such sources if fewer exist. The reservations are disjoint. At least $N - 5m \ge 7m + 8 \ge 15$ sources remain when $m \ge 1$; when $m = 0$, none are reserved and $N \ge 24$. The corner theorem supplies a target on the unreserved sources. Orient its possible monochromatic edge along the shortest direction. If the target repeats colour r adjacently, a compatible source with adjacent r-faces survived the reservation, so the r-reserve has its full size m. It fills the monochromatic edge. The Hall observation completes the frame.

One rare pair $\alpha\beta$. For each of the three ring colours, reserve up to m sources in the whole collection having that colour on adjacent faces, again taking all such sources if fewer exist. At most $3m$ family copies are removed, so $\mathcal{H}_{\alpha\beta}$ retains at least

$$(3\tau - M + 3) - 3m = 3t + 2M + 3 \ge 8$$

copies, and at least eleven total sources remain unreserved. Apply Lemma 11 to eight unreserved family copies and any three further unreserved sources. The resulting corner target lies in $\mathcal{H}_{\alpha\beta}$. Orient its $\alpha\beta$ axis along the shortest direction. Any monochromatic edge is parallel to this axis and has its full reserve of m copies, by the same survivor argument. The target avoids the only rare pair, so the Hall observation applies.

Two rare pairs e, f. Their common missing variety V is the unique opposite-repeat variety with e, f as singleton opposite pairs. Its multiplicity satisfies

$$v \geq N - 2(d-1) = 2(m+t) - 2 \geq 2.$$

If $v \geq 8$, use eight copies for the corners of V. If $4 \leq v \leq 7$, the family $\mathcal{H}_e$ has at least eleven copies and hence at least four not of variety V. Each shares at least four corners with V, by the ring array. Place four such copies at distinct corners and fill the other four with V. In either case, V has no monochromatic edge and avoids both rare pairs; the Hall observation applies.

It remains that $v = 2$ or 3, forcing $m + t = 2$ and $N = 2d$. If n_e, n_f count the sources possessing the rare pairs, then

$$n_e, n_f \leq d - 1, \qquad v \geq N - n_e - n_f.$$

Consequently,

$$n_e, n_f \geq d - 2, \qquad n_f + v \geq d + 1.$$

Use exactly the reservation and the construction of Lemma 11 from the one-rare-pair case to obtain a corner-and-monochromatic-edge matching for a target $T \in \mathcal{H}_e$, with its e axis shortest.

Within this family, V shares at least four target corners and can miss at most two geometric edges, both parallel to the e axis. For a different opposite-repeat target these are its two f-edges; for an adjacent-repeat target they are its f-edge and monochromatic edge. Thus V misses at most $2m + 4 \leq 6$ actual positions.

The only Hall sets not covered as before contain bichromatic edge positions of type f and no non-rare bichromatic edge. Such a set D has at least n_f neighbours. If $|D| > n_f \geq d - 2 \geq 8$, every copy of V is also a neighbour, since it misses at most six positions. These are distinct from the n_f copies possessing f, so

$$|\Gamma(D)| \geq n_f + v \geq d + 1.$$

This handles $|D| \leq d$; larger sets neighbour every source. Hall's theorem completes the frame in the final case.

The number of frame positions gives the lower bound, proving equality. All preliminary placements certify Hall inequalities; the final matching may rearrange them. □

6.6 Completion of the classification

Proof of Theorem 1. Order the sides as $2 \leq a \leq b \leq c$. When $a + b \geq 6$, Theorem 33 gives $F_5 = L$. The remaining possibilities are exactly $(a, b) = (2, 2)$ and $(2, 3)$, settled by Proposition 31 and Theorem 26. Together with the corner theorem at $(2, 2, 2)$, these give $F_5 = 15$ for sums six and seven, $F_5 = 17$ for sum eight, and $F_5 = L$ for every larger sum. Proposition 22 proves sharpness and yields the stated maximum formula.

For the complete box, take $\max\{abc, F_5(a, b, c)\}$ sources and first choose a frame. Every source has every colour, so any unused source can fill any face-interior position after a suitable proper rotation. All fully interior positions are unrestricted. There are enough unused sources to fill the remaining $abc - L$ positions. Conversely, any complete box supplies a frame and uses abc sources. Hence

$$G_5(a, b, c) = \max\{abc, F_5(a, b, c)\}.$$

Putting $x = a - 2$, $y = b - 2$, and $z = c - 2$ gives

$$abc - L = xyz + 2(xy + xz + yz) \geq 0,$$

which simplifies this identity to the claimed formula for G_5.

If $(a, b) = (2, 2)$, this formula is $\max\{4c, 15, 2c+9\}$: it exceeds the volume only for $c = 2, 3, 4$. In every other sorted case, $b \geq 3$; then $abc \geq 18 \geq 15$ and

$$abc - 2(a + b + c) - 1 \geq 2bc - 2b - 2c - 5 \geq 1.$$

The first inequality follows by increasing a from two, and the second is minimised at $b = c = 3$. There are no further exceptions. □

7 Constructing an assembly

The threshold theorem guarantees some target colouring, which need not be specified in advance. It can be found by a bounded number of capacitated matching problems, even when the side lengths are large.

Proposition 34 (Compressed construction)**.** *Given an explicit list of q sources and $a, b, c \geq 2$, one can find a frame, or report that none exists, in $O(q + L)$ time, where $L = 4(a + b + c) - 16$. A complete-box assembly can be found, or ruled out, in $O(q + abc)$ time. These bounds use the word-RAM model with the input sizes, dimensions and output indices fitting in a word. On collections covered by the corresponding threshold, the running time is $O(q)$. For sources given as 75 binary multiplicities, feasibility and a compressed assembly have polynomial bit complexity; expanding the placements costs the output size.*

Proof. Classify each source among the 75 varieties by testing its 24 proper rotations, recording both its variety and a rotation to the representative. This takes $O(q)$ time and gives 75 multiplicities and lists of actual copies.

Test each target variety with each distinct assignment of the three side lengths to its coordinate axes. There are at most six assignments, hence at most 450 configurations. When exactly two side lengths agree, three assignments suffice, giving 225 configurations; when all three agree, only 75 are needed. Every physical target is a proper rotation of one of the 75 representatives, and every proper rotation permutes the three coordinate axes. Bringing the target to its representative therefore permutes the side-length assignment. Sign changes only reverse positions along an edge; they introduce no additional demand pattern. This proves that the listed configurations cover every assembly.

For each configuration, use a flow network with 75 source-variety nodes and at most twenty requirement nodes: eight corners of demand one and twelve geometric edges whose demands are their assigned side lengths minus two. Omit zero demands. Add an arc from the network source to each variety with capacity its multiplicity, and an arc from each requirement to the sink with capacity its demand. Connect a variety to a requirement exactly when it fits, with capacity q. There are at most 97 vertices, including source and sink. Identical requirements may be merged.

An integral flow of value L saturates every requirement and gives the number of copies of each variety to place there. Conversely, any frame induces such a flow. A shortest-augmenting-path max-flow algorithm has a number of arithmetic operations bounded in terms of the network size, independently of the capacities [4]. Here the network size and the number of configurations are constants. Thus all flow tests take $O(1)$ word operations, or polynomial time in the bit lengths for binary input. If every test fails, no frame exists.

Expand a successful flow into L distinct source copies and positions. A precomputed proper rotation for each compatible corner or edge, composed with the recorded source rotation, gives each placement's orientation. This takes $O(L)$ time. A complete box exists exactly when a frame exists and $q \geq abc$: fill face interiors with unused copies rotated to expose the required single colour, and fill the interior arbitrarily. Writing these placements takes $O(abc)$ time. □

8 Collections with few varieties

Let $M(k)$ be the least number of cubes guaranteeing a corner solution when the collection contains at most k varieties.

Lemma 35. *Two five-colour varieties with disjoint corner sets can build an opposite-repeat target from four copies of each.*

Proof. This is a small finite geometric check. Of the $\binom{75}{2}$ unordered pairs, 330 have disjoint corner sets. Up to colour permutation and exchange of the sources, they have the following five representatives. The first source is $(0,1,0,2,3,4)$ throughout; tuples use the face order of Section 2. Each listed target has four corners supplied by each source.

Second source	Target
$(0,1,1,2,4,3)$	$(0,0,1,2,3,4)$
$(0,1,2,3,2,4)$	$(0,1,2,2,3,4)$
$(0,1,2,3,3,4)$	$(0,1,2,2,3,4)$
$(0,1,2,4,3,4)$	$(0,1,2,2,3,4)$
$(0,3,0,4,1,2)$	$(0,0,1,2,3,4)$

The accompanying program `src/five_colour_extensions.cpp` regenerates the varieties and checks the orbit coverage and all five witnesses. □

Theorem 36. *For every integer $k \geq 1$,*

$$M(k) = \begin{cases} 8 & k = 1, \\ 13 & k = 2, \\ 15 & k \geq 3. \end{cases}$$

Consequently, three is the minimum number of varieties in a fourteen-cube collection without a solution.

Proof. Eight identical copies build their own variety, and fewer than eight pieces cannot form a corner solution. For two varieties, consider thirteen copies. If either variety occurs at least eight times, it builds itself. Otherwise the multiplicities are seven and six. If the varieties share a corner type, use a minority copy at that corner of the majority target and seven majority copies elsewhere. If their corner sets are disjoint, apply Lemma 35.

For the lower bound take six copies each of

$$U = (0,1,0,2,3,4), \qquad Q = (0,1,1,3,2,4),$$

where U is the variety of Lemma 21. Their corner sets share just 041. Inspection of the 75 varieties shows that only U, Q have all corner types in their union. Each has seven private positions, so neither can be built with six copies. The same geometric checker verifies these assertions. Thus $M(2) = 13$. For $k \geq 3$, Theorem 3 supplies the upper bound, and the collection in Lemma 12 supplies the lower bound. □

Remark (A family of fourteen-cube obstructions). Keep A, C from Section 4, and put

$$B_1 = (0,1,2,3,2,4), \quad B_2 = (0,1,2,3,3,4), \quad B_3 = (0,1,2,4,3,4).$$

All 21 collections

$$4A + 5C + x_1B_1 + x_2B_2 + x_3B_3, \qquad x_1 + x_2 + x_3 = 5, \quad x_i \in \mathbb{Z}_{\geq 0},$$

are unsolvable. The B_i share the six tricoloured corner types $023, 034, 042, 124, 132, 143$. The combined corner union allows only A, C, B_1, B_2, B_3 as targets, as a check of the catalogue confirms. Target A needs five A-copies, C needs six C-copies, and any B_i needs six copies from the B-group. The respective supplies are four, five and five, regardless of their distribution within the B-group. The original example is $x_1 = 5, x_2 = x_3 = 0$.

9 Separate formal verification

The threshold statements have a Lean 4 and mathlib verification, both at `v4.35.0`, using the physical model and palette-independent geometry of `colour-cubes-core` [6]. This verifies the same conclusions through a different corner proof: physical cubes are connected to catalogue matchings, a complete finite root cover, and checked exclusion certificates. It does not formalise the new elementary derivation or every lemma of the mathematical appendix. The cuboid deductions then apply the verified corner theorem.

In namespace `FiveColour.Puzzle`, `Solution.lean` exports the unconditional `box_frame_threshold` and `complete_box_threshold`. They use the project's physical box definitions and proper-rotation bridges. The separate Mathlib-only comparator challenge covers the corner problem; a full external comparator pass remains unconfirmed. The construction algorithm's complexity bound is a mathematical deduction, not a Lean claim.

The accompanying *Certificate proof and verification supplement* retains the alternative mathematical reduction, interval-checker soundness, implementation details and reproduction commands. This supplement and the repository explain the formal route; they are not premises of the elementary proof. The command `make verify-full` replays the projected certificates, builds the unconditional cuboid results and audits their dependencies. The shorter `make check` does not reproduce the full theorem.

10 Further structure and questions

The main thresholds leave questions about the structure of extremal obstructions. A separate computer-assisted classification identifies 11,430 failing fourteen-source inventories in three families. All have the invariant

$$3\sum_j \binom{g_j}{2} - 4\binom{o}{2} = 36,$$

where g_j counts copies in an adjacent class and o is the total number of opposite copies. This gives a different proof of the corner threshold by single-copy deletion. Its new Lean replay is still in progress; these classification claims are not included in the formal-verification statement above or used in the elementary proof. An elementary classification of all fourteen-source failures remains open, as does finding a substantially shorter proof of the threshold. Boxes with a side of length one require a different exposure model and are outside the classification proved here.

AI disclosure and review status

GPT-6-Astra (OpenAI), used through Codex, developed the arguments, search and verification programs, Lean formalisation and manuscript under the human author's direction. The author initiated and directed the research. Author review and submission approval of this working manuscript are pending. The inherited geometry has its own provenance in the pinned dependency. Automated checks and AI review are not independent human peer review.

References

[1] P. A. MacMahon, *New Mathematical Pastimes*, Cambridge University Press (1921). Digitised edition.

[2] E. Berkove, A. Eickemeyer and M. Schulman, *The 4-Color Cubes Puzzle*, Integers **18** (2018), A68. doi:10.5281/zenodo.10680216. Journal PDF.

[3] P. Hall, *On Representatives of Subsets*, Journal of the London Mathematical Society **s1-10**(1) (1935), 26–30. doi:10.1112/jlms/s1-10.37.26.

[4] J. Edmonds and R. M. Karp, *Theoretical Improvements in Algorithmic Efficiency for Network Flow Problems*, Journal of the ACM **19**(2) (1972), 248–264. doi:10.1145/321694.321699.

[5] J. Davies, *Three-colour solutions from four-colour cubes*, manuscript (2026). Accompanying repository.

[6] J. Davies, `colour-cubes-core`: palette-independent Lean definitions and proofs for face-coloured cubes, Lean library (2026). Release `v1.0.0`, `https://github.com/jasondavies/colour-cubes-core`.

[7] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in Automated Deduction—CADE 28, LNCS **12699**, Springer (2021), 625–635. doi:10.1007/978-3-030-79876-5_37.

[8] The mathlib Community, *The Lean Mathematical Library*, in Proceedings of CPP 2020, ACM (2020), 367–381. doi:10.1145/3372885.3373824. arXiv:1910.09336.